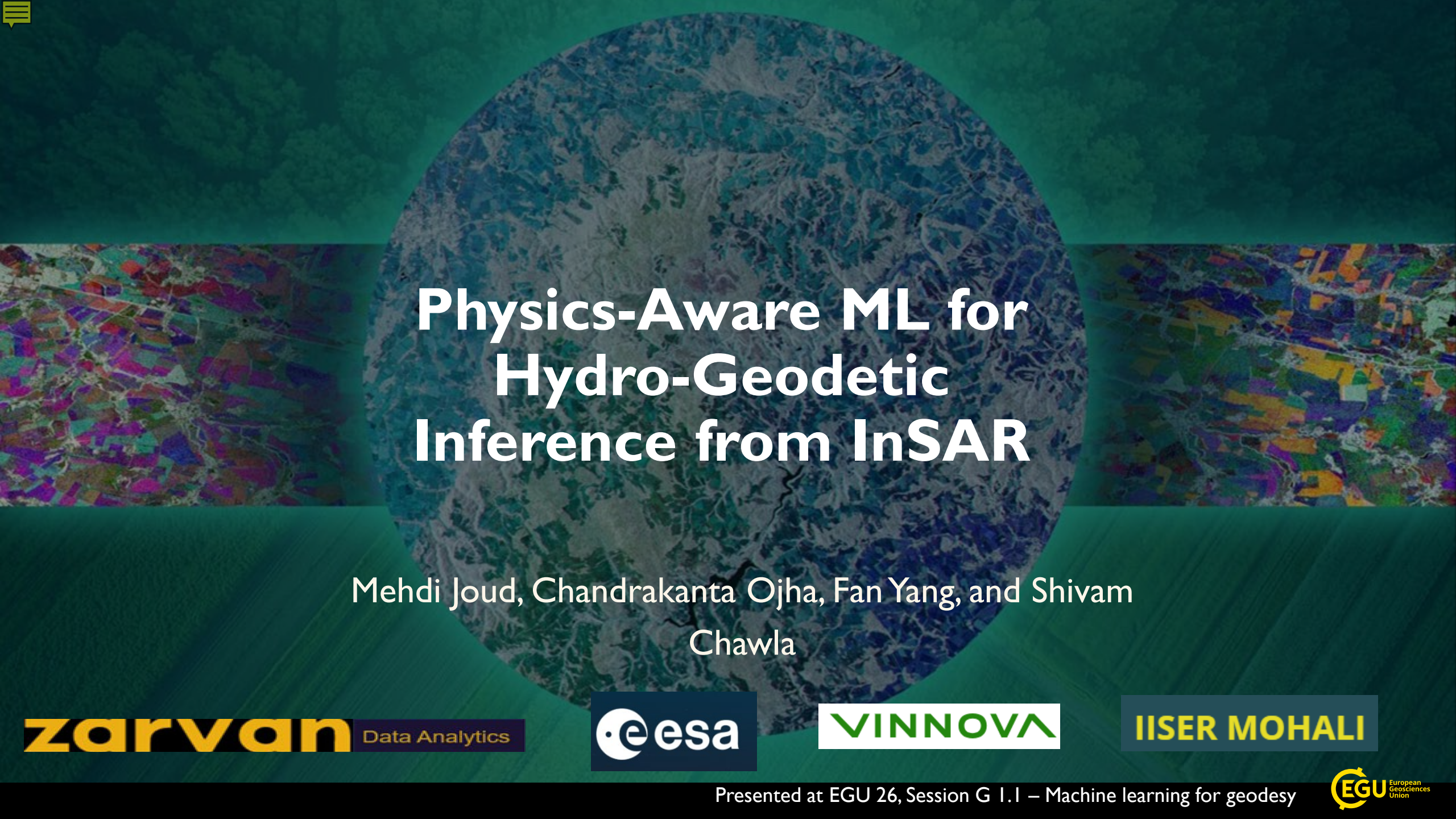




Physics-Aware ML for Hydro-Geodetic Inference from InSAR

Mehdi Joud, Chandrakanta Ojha, Fan Yang, and Shivam
Chawla

zarvan Data Analytics



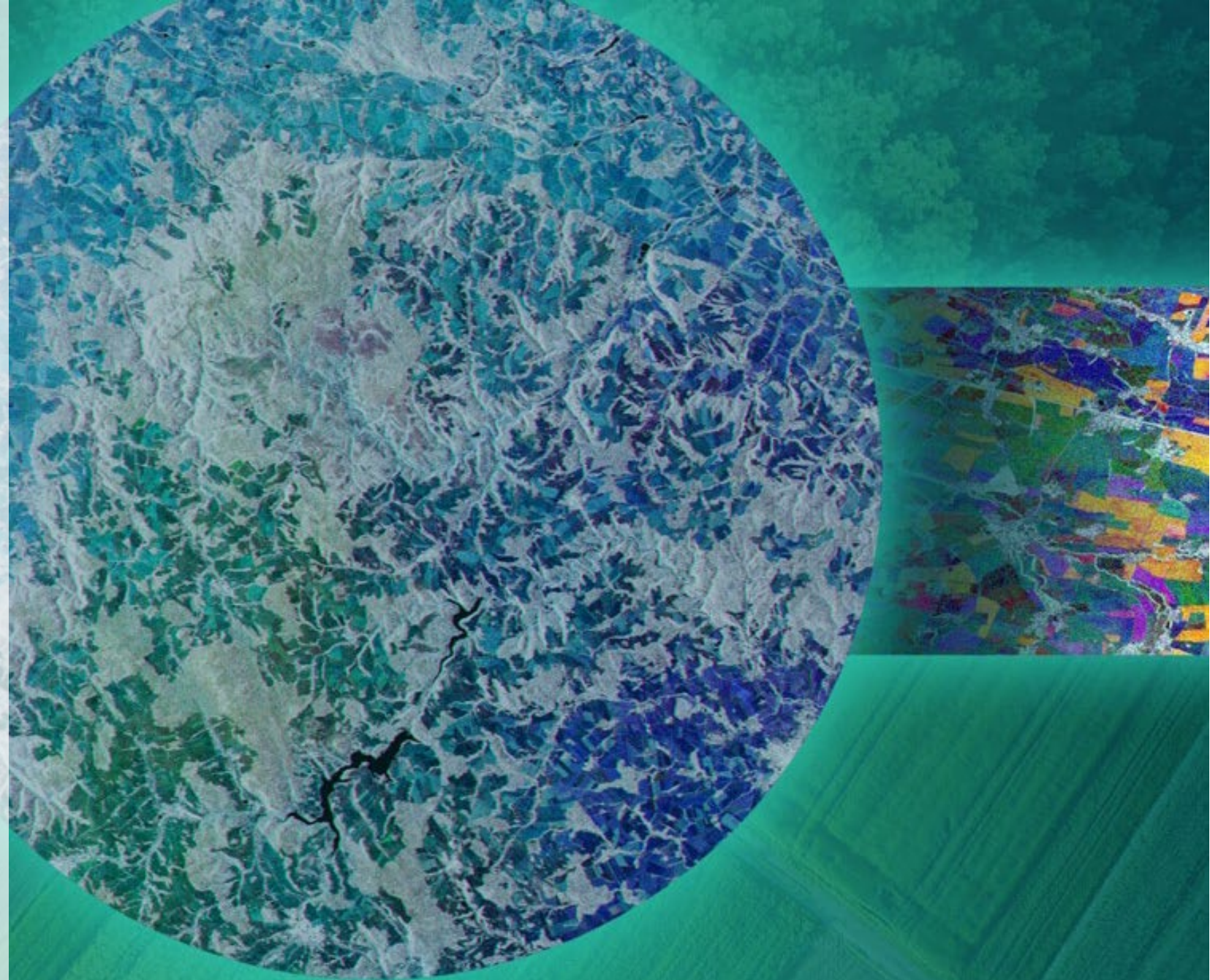
VINNOVA

IISER MOHALI

Presented at EGU 26, Session G I.I – Machine learning for geodesy



Physics-Aware Machine Learning for Large-Scale Hydro-Geodetic Inference from InSAR





The problem: Aquifer-Scale Hydrology Is Missing

Hydrology models & gravimetry are coarse; aquifer processes occur at km–sub-km scale.



InSAR Observations: The Surface Signal as Proxy for Subsurface Water Storage

InSAR observes pressure-driven deformation linked to groundwater changes.





The Inversion Gap: Ill-Posed, Vertically Resolved Hydrology Inversion

Multi-decadal, multi-mission InSAR
→ nonlinear, unstable inversion
problem.



Hybrid Bayesian + Physics-Aware Transformer Framework

InSAR time series → spatio-
temporal cubes → ML inversion.



Physics-Aware Learning

InSAR time series → spatio-temporal cubes → ML inversion.

$$u(x, y, t) = \sum_k G_k S_k(x, y, t)$$

where G_k are Green's functions describing elastic and poroelastic deformation for each layer.

$$\hat{S}(x, y, t) = F_\theta(u(x, y, t))$$

The model learns spatial and temporal deformation patterns across the region — not pointwise, but as a coupled system.”

$$\hat{u} = \sum_k G_k \hat{S}_k$$

We penalize the mismatch:

$$L_{\text{physics}} = \|u - \hat{u}\|^2$$

We also enforce basin-scale mass conservation using gravimetric constraints:

$$L_{\text{mass}} = (\text{mean}(\hat{S}) - \text{TWS})^2$$

So the ML model operates **within geodetic physical laws.**”

Reserved slide I: What Makes It Physics-Aware

Physical Constraints in Learning

Elastic & poroelastic
forward operators + basin-
scale mass conservation +
structured supervision.

- A Swin U-Net predicts $\hat{S}_g$ from inputs X (InSAR u_z + auxiliaries).
- You still use the same Green's kernel G —but now in the forward direction:

$$\hat{u} = G * (\rho_w g \hat{S}_g)$$

- Train the network by minimizing a sum of losses:
 - **Data term** (if labels exist): $L_{\text{MSE}} = \|\hat{S}_g - S_g^{\text{ref}}\|^2$
 - **Physics term**: $L_{\text{green}} = \|u_z - \hat{u}\|^2$ (enforces Biot consistency)
 - **Mass term (GRACE)**: $L_{\text{GRACE}} = (\text{mean}_{\Omega}(\hat{S}_g) - TWS)^2$
 - (optional) smoothness L_{reg}

After training, inference is one pass through the network: $X \rightarrow \hat{S}_g$.

Algorithm 2 Physics-Informed AI with Multi-Source Supervision

Require: Inputs $X[t, y, x, c]$ with $c = \{u_z, S_0, \dots\}$; reference $S_g^{\text{ref}}[t, y, x]$ (if available); GRACE $TWS[t]$; compact Green kernel G ; ρ_w, g ; weights $\lambda_G, \lambda_T, \lambda_R$; optimizer (AdamW)

Ensure: Trained θ ; predictions $\hat{S}_g[t, y, x]$ consistent with physics and multi-scale mass

- 1: Precompute $\hat{G} = \text{FFT2}(G)$
- 2: **for** epochs **do**
- 3: **for** mini-batch $(X_b, S_{g,b}^{\text{ref}}, TWS_b)$ **do**
- 4: **Forward:** $\hat{S}_{g,b} \leftarrow F_{\theta}(X_b)$
- 5: **Physics reconstruction in deformation domain:**
 $\hat{u}_b \leftarrow \text{IFFT2}(\hat{G} \cdot \text{FFT2}(\hat{S}_{g,b} \rho_w g))$
- 6: **Loss terms:**
 - $L_{\text{MSE}} = \|\hat{S}_{g,b} - S_{g,b}^{\text{ref}}\|_2^2$ ▷ supervision, if labels exist
 - $L_{\text{green}} = \|u_{z,b} - \hat{u}_b\|_2^2$ ▷ physics consistency
 - $L_{\text{GRACE}} = (\text{mean}_{\Omega}(\hat{S}_{g,b}) - TWS_b)^2$ ▷ basin-scale mass
 - $L_{\text{reg}} = \|\nabla \hat{S}_{g,b}\|_2^2$ ▷ spatial smoothness, optional
- 7: **Total loss:** $L = L_{\text{MSE}} + \lambda_G L_{\text{green}} + \lambda_T L_{\text{GRACE}} + \lambda_R L_{\text{reg}}$
- 8: Backpropagate $\nabla_{\theta} L$ and update θ (AdamW)
- 9: **end for**
- 10: **end for**
- 11: **Inference:** given new X , output $\hat{S}_g = F_{\theta}(X)$
- 12: **return** $\theta, \hat{S}_g$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla_{\theta} L,$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla_{\theta} L)^2,$$

$$\theta \leftarrow \theta - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \eta \lambda_w \theta$$

Mathematically

1. Sample a mini-batch $\{(X_i, y_i)\}_{i=1}^B$.
2. Forward: $\hat{y}_i = F_{\theta}(X_i)$.
3. Compute $L(\theta) = \frac{1}{B} \sum_i \ell(\hat{y}_i, y_i)$.
4. Backprop: compute gradients $\nabla_{\theta} L$.
5. Optimizer update (AdamW):

One training step

```
with autocast(): # mixed precision matmuls (speed, less memory)
    pred = model(X) # forward
    loss = criterion(pred, y) # MSE + TV
optimizer.zero_grad()
scaler.scale(loss).backward() # backprop in fp16-safe way
scaler.step(optimizer) # AdamW update
scaler.update()
```

Reserved slide 2: The Model

3D Swin Transformer U-Net

- **Multi-Mission, Multi-Decadal Batch Processing**

Massive InSAR archives + GPU-based ML inversion → needs HPC.

- Patch embedding: spatio-temporal cubes $(h \times w \times \tau) \rightarrow$ tokens
- Windowed attention: in each window W ,
$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V$$
- Shifted windows: shift windows next layer → cross-window mixing (global coverage, local cost)
- Encoder–Decoder: hierarchical encoder (patch merge) + decoder (upsample) with skips
- Output: $\hat{S}_g(t, y, x)$ — predicted groundwater storage map(s)

<https://github.com/microsoft/Swin-Transformer>

Liu, Z., et al. 2021. In *Proceed. IEEE/CVF international conference on computer vision*, pp. 10012-10022. 2021.

<https://github.com/microsoft/aurora/tree/main>

Bodnar, C., et al. 2025. A foundation model for the Earth system. *Nature*, pp.1-8.

INPUT

$X \in \mathbb{R}^{T \times H \times W \times C_{in}}$
where $C_{in} = 1$ (InSAR).

Network prediction:

$\hat{Y} = F_{\theta}(X)$,
where F_{θ} is the Swin U-Net,
, with trainable weights θ .

$$\mathcal{L}(\hat{Y}, Y) = \frac{1}{THWC_{out}} \sum |\hat{Y} - Y|$$

$$\mathcal{L} = \|\hat{S}_g - S_g^{ref}\|_2^2 + \lambda_R \|\nabla \hat{S}_g\|_2^2$$

OUTPUT

$Y \in \mathbb{R}^{T \times H \times W \times C_{out}}$,
where $C_{out} = 3$ (Sg, Sd, S0).

$$\text{softmax}(z)_j = \frac{e^{z_j}}{\sum_{\ell} e^{z_{\ell}}}, \quad z_j = (q \cdot k_j) / \sqrt{d}.$$



Model and Data Fusion

The Model

3D Swin Transformer U-Net

**Multi-Mission, Multi-Decadal
Batch Processing**

Massive InSAR archives + GPU-
based ML inversion → needs HPC.



Main Result: Grouped Posterior

The Model

3D Swin Transformer U-Net

**Multi-Mission, Multi-Decadal
Batch Processing**

Massive InSAR archives + GPU-
based ML inversion → needs HPC.



Independent Validation

The Model

3D Swin Transformer U-Net

**Multi-Mission, Multi-Decadal
Batch Processing**

Massive InSAR archives + GPU-
based ML inversion → needs HPC.

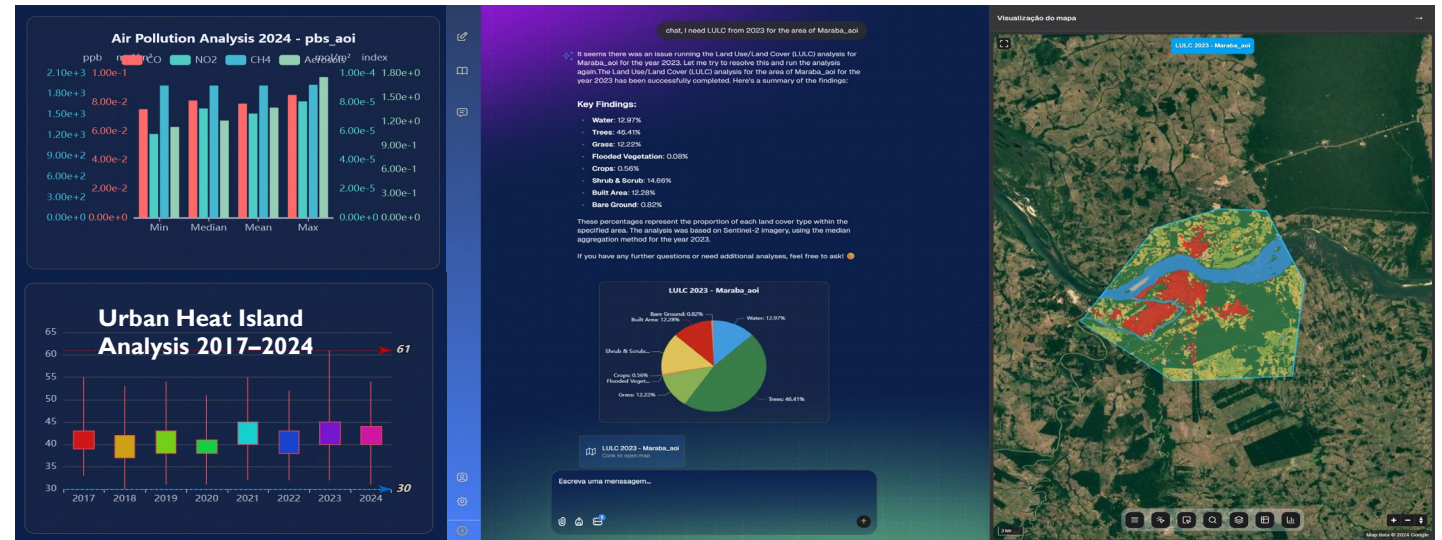
From Research to AID/AIT

The Model

3D Swin Transformer U-Net

**Multi-Mission, Multi-Decadal
Batch Processing**

Massive InSAR archives + GPU-
based ML inversion → needs HPC.



Conclusions

Key Takeaway and proposals

Geodesy + Physics + ML + HPC

ML extends classical inversion, does not replace physics.

What This Enables

Aquifer-scale hydrology, model improvement, and groundwater-related geohazard insight..